

Fixed-Degree Smoothness Barriers for Real Quadratic Infrastructure and the Joux–Buchmann Bridge

Abstract. The Joux/BGJT algorithm achieves quasi-polynomial complexity for discrete logarithms in small-characteristic function fields by exploiting Frobenius symmetry, tower descent, and polynomial-basis smoothness. We investigate whether a structural analogue exists for the real quadratic regulator of $K = \mathbb{Q}(\sqrt{N})$, which would yield a faster classical algorithm for computing R_K and hence for factoring RSA semiprimes $N = pq$. We prove two barriers. First (Theorem 1), in the class of Buchmann-style smooth-relation algorithms for fixed real quadratic degree, the heuristic relation-collection barrier is $L_D[1/2]$, conditional only on the standard smoothness heuristic and even relative to an integer factoring oracle. Second (Theorem 2), in the generic infrastructure model, finding the infrastructure period requires $\Omega(\sqrt{R_K})$ oracle queries. The Joux pipeline analysis shows Gate IV failure: no Frobenius analogue, no tower descent, and no sub- $L[1/2]$ mechanism exists in degree 2. The conclusion is a precise corollary: any Joux-style improvement would require either a Frobenius analogue not known to exist, or a semiprime-specific prefix-to-period law for continued fractions of $\sqrt{pq}$.

1. Introduction

The Number Field Sieve achieves $L_N[1/3]$ for integer factoring. In 2013, Joux and independently Barbulescu–Gaudry–Joux–Thomé showed that discrete logarithms in $\mathbb{F}_{p^n}$ for small characteristic p can be solved in quasi-polynomial time $L[1/4+o(1)]$ [Joux 2013, BGJT 2014]. The key ingredients — Frobenius symmetry, polynomial smoothness, and recursive tower descent — collectively break the $L[1/2]$ barrier.

The real quadratic regulator R_K of $K = \mathbb{Q}(\sqrt{D})$ governs the fundamental unit ε_D and the period of the continued fraction expansion of $\sqrt{D}$. For $D = N = pq$, factoring and regulator computation are empirically and conjecturally tightly linked (the BPW bridge; see companion Paper B). If the regulator could be computed sub- $L[1/2]$, this would likely give a sub-NFS factoring algorithm.

This paper asks: *Can the Joux/BGJT approach transfer to real quadratic regulators?*

The answer is no. We identify the five structural components of the Joux method (§2), analyze their transfer (§3), and prove two barrier theorems (§4–5). The analysis is conducted as a pipeline (“gate test”), and Gate IV — the test for an actual sub- $L[1/2]$ mechanism — fails hard.

Limits of claims. This paper does not prove that no classical algorithm fac-

tors N in sub- $L[1/3]$ time. It proves that *this specific class of approaches* — smooth-relation regulator algorithms and infrastructure period-finding — cannot break $L[1/2]$ within the stated models. Theorem 2 is model-relative; it does not exclude non-black-box algorithms exploiting specific arithmetic properties outside the generic infrastructure model.

Standing assumptions. The Dickman–CEP smoothness heuristic is used for Theorem 1; it is unproven but standard. Theorem 2 holds unconditionally in the generic infrastructure model. GRH is not needed for the main barrier results.

2. The Joux/BGJT algorithm: five components

The Joux/BGJT method decomposes into five structural components:

ID	Component	Role in BGJT
K1	Smoothness basis	Low-degree polynomials generating many relations cheaply
K2	On-the-fly relations	Target-specific degree-reducing elimination
K3	Frobenius symmetry	Large fixed-locus identity; $X^q \equiv \prod_{a \in \mathbb{F}_q} (X - a)$
K4	Tower/subfield descent	Recursive halving of effective degree
K5	Representation switching	Cheap transition between polynomial and rational models

K3 and K4 are the heart of the $L[1/4]$ gain. Without them, the algorithm would not break $L[1/2]$.

3. Transfer analysis to real quadratic fields

K1 (smoothness basis): Real quadratic fields admit factor bases of prime ideals of bounded norm. The analogue is workable in principle. *Status: partial pass.*

K2 (on-the-fly relations): Principal ideal relations exist and can be collected. However, no norm-shrinking identity analogous to $X^q \equiv h_0/h_1$ is known. Relations cannot be recycled via a target-specific descent step. *Status: operationally yellow, functionally red for the BGJT mechanism.*

K3 (Frobenius symmetry): The intrinsic automorphism group is $\text{Aut}_{\mathbb{Q}}(K) = \{1, \iota\}$ where $\iota(\sqrt{N}) = -\sqrt{N}$. This two-element group cannot supply a large fixed locus or a Frobenius-type splitting identity. The degree-2 constraint is fundamental: real quadratic fields have no intermediate subfields, so no iterated Frobenius collapses degree. *Status: fail.*

K4 (tower descent): No intermediate subfields exist. External tower constructions fail due to discriminant growth, class-field data requirements, or loss of connection to R_K . The regulator does not factorize under tower extension in a norm-decreasing way. *Status: fail.*

K5 (representation switching): Multiple representations of the infrastructure exist (ideals, binary quadratic forms, NUCOMP/NUDUPL, Arakelov divisors). In fixed degree 2, switching between them does not create the height/degree balance that enables $L[1/3]$ -type descent in number fields. *Status: yellow for flexibility, red for asymptotic gain.*

4. Gate analysis

Gate	Test	Result	Reason
Gate I	K1–K5 vocabulary coherent in principle	PASS	Factor bases and relation collection transferable
Gate II	Weak analogues of K2/K5 exist	PASS	Principal ideal relations and representation switching present
Gate III	Replacement mechanisms formulable	PASS	Partial replacements articulable
Gate IV	Some pipeline achieves sub- $L[1/2]$	FAIL	All pipelines collapse to $L[1/2]$ or worse

5. Barrier theorems

Theorem 1 — Fixed-degree norm-smoothness barrier

In the class of Buchmann-style regulator algorithms for fixed real quadratic degree, based on smooth principal-ideal relations, the optimal heuristic relation-collection complexity is $L_D[1/2, O(1)]$, even relative to an integer factoring oracle. The factoring oracle accelerates norm factorization but does not alter the

probability that candidate norms are smooth. Hence the $L_D[1/2]$ bottleneck is structural, not arithmetic.

Proof sketch. Set the factor base size $B = L_D[a, c]$ and let $X = D^{O(1)}$ be the size of candidate relation norms, so $\log X \asymp \log D$. The smoothness parameter is

$$u = \frac{\log X}{\log B} \asymp \frac{\log D}{(\log D)^a (\log \log D)^{1-a}} = \frac{(\log D)^{1-a}}{(\log \log D)^{1-a}}.$$

By the Dickman–CEP heuristic the inverse smoothness probability is $\rho(u)^{-1} = L_D[1-a, O(1)]$. Collecting $L_D[a, O(1)]$ independent relations (one per factor-base element) at cost $\rho(u)^{-1}$ each gives total relation-collection cost

$$L_D[a, O(1)] \cdot L_D[1-a, O(1)] = L_D[\max(a, 1-a), O(1)].$$

This is minimized at $a = 1/2$, giving $L_D[1/2, O(1)]$.

A factoring oracle replaces the per-relation norm factorization step (cost $L_D[0]$) with essentially free factoring, but does not change the *probability* that a given candidate norm is B -smooth — that depends only on the size of X and B . Hence u and $\rho(u)$ are unchanged, and the balance point $a = 1/2$ persists. $\square$

Theorem 2 — Infrastructure period barrier (generic model)

In the generic infrastructure model — where algorithms evaluate the group law and distance function as black-box oracles but have no access to internal arithmetic structure — any classical algorithm finding the infrastructure period R_K by distance-testing requires $\Omega(\sqrt{R_K})$ oracle queries. Baby-step/giant-step methods achieve $O(\sqrt{R_K})$ queries and are optimal within this model.

Note. This is a model-relative lower bound, analogous to the Generic Group Model lower bound for DLP. It does not exclude algorithms that exploit specific arithmetic properties of the real quadratic infrastructure — for example, algorithms based on lattice reduction, Arakelov theory, or number-theoretic properties of the continued fraction expansion of $\sqrt{N}$ — that fall outside the black-box oracle model.

6. Conclusion and corollary

Corollary. A Joux-style breakthrough below $L_D[1/2]$ for real quadratic regulators requires a mechanism outside both the smooth-relation model (Theorem 1) and the generic infrastructure model (Theorem 2). Concretely: either a non-factorization-dependent Frobenius analogue, or a new semiprime-specific prefix-to-period law for continued fractions of $\sqrt{pq}$.

No such mechanism is known. The Joux/BGJT bridge to real quadratic factorization is closed within the studied models.

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